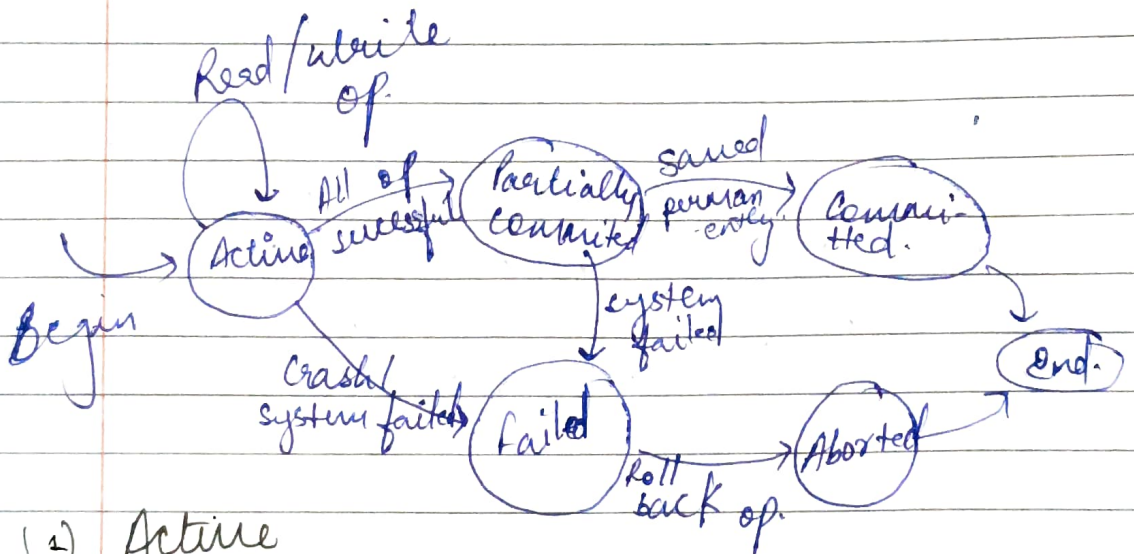


Unit - 3

* States of transaction

- 1) Active state
- 2) Partially committed
- 3) Failed -
- 4) Committed
- 5) Aborted (6) End -



(1) Active

In this state transaction is being executed & this is the initial state of every tx.

2) Partially committed when transaction execute its final operation, then it is said P.C.

3) Failed.

A transaction is said to be in a failed state if any of the check made by the database recovery system failed. A failed transaction can no longer be processed further.

(4) Aborted - If any of the check fails and the transaction has reached state then the recovery manager roll back all its ~~write~~^{write} operation on the database to bring the data base back to its original state.

5) Committed -

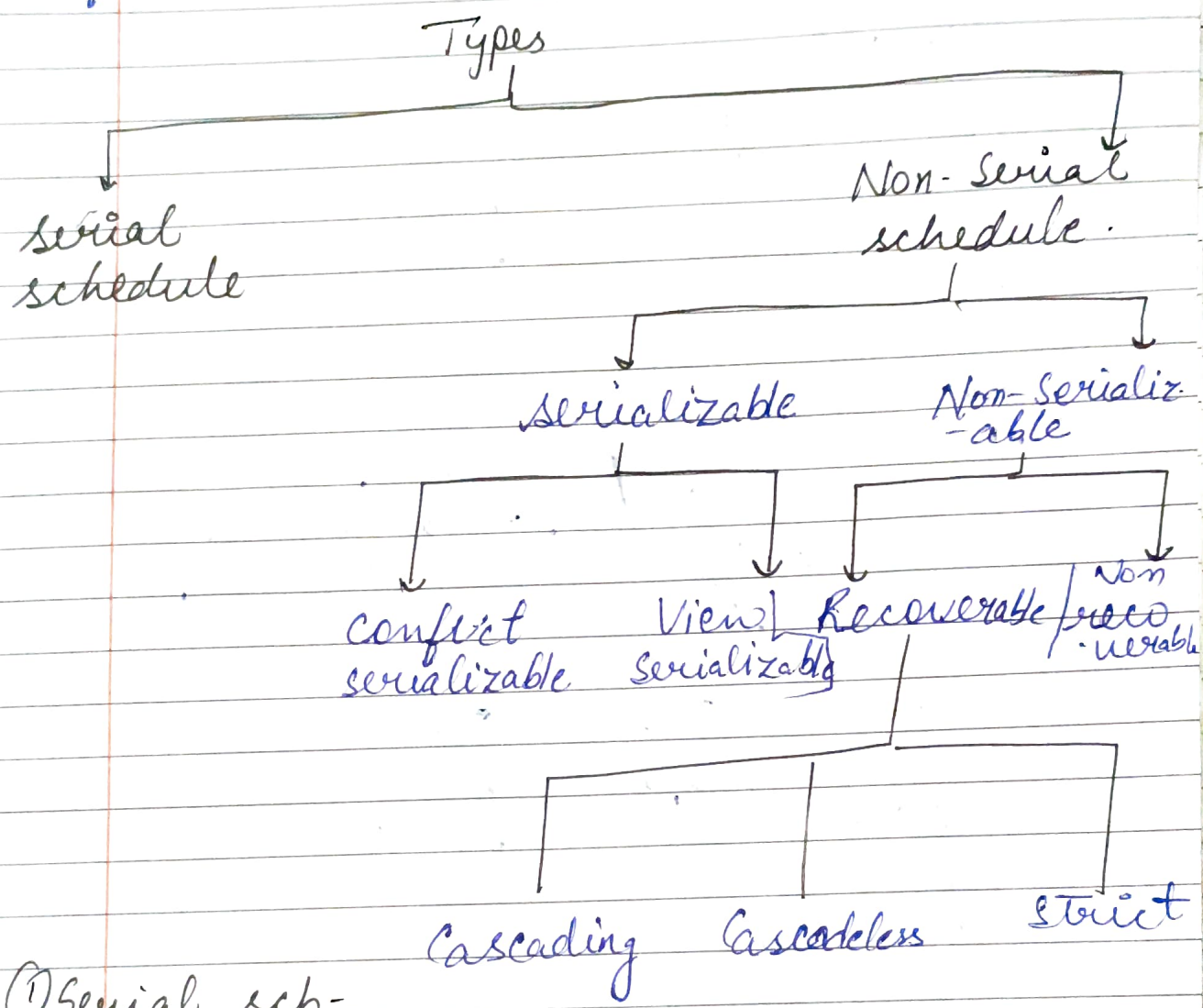
If a transaction execute all its operation successfully then it is said to be in committed state. And all the changes are now permanently saved in the system.

* diff transaction operation -

- 1 begin-transaction → done before active state
- 2 Read-item and write-item → At active state
- 3 end-transaction → at the end
- 4 commit → after partially committed.
- 5 Roll back → before abort

* Schedules in DBMS-

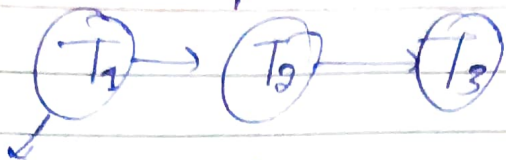
A schedule is an order in which the operation ~~are~~ transaction appear of multiple for the execution.



① Serial sch-

The sch in which transaction are executed non interleaved that is a serial sch is one in which non transaction starts running until a running transaction has ended

In serial T_1, T_2 & T_3 are
 diag. independent to each other.



once all the op of T_1 executed when T_2 starts.

T_1	T_2	T_3
R(A)		
W(A)		
R(B)		
WR(B)		
	R(A)	
	W(A)	
		R(C)
		W(C)

(2) Non serial sch -

In which op of multiple transac-
 -ion are interleaved and this
 this might lead to concurrency
problem.

occur in
 Non serial
 sch

T_1	T_2	T_3
R(A)	R(C)	R(X)
W(A)		
R(B)	W(C)	W(X)
W(B)		

T_1	T_2	T_3
R(A)		
W(A)	R(B)	
	W(B)	
R(A)	W(A)	R(Y)

Serializability in DBMS

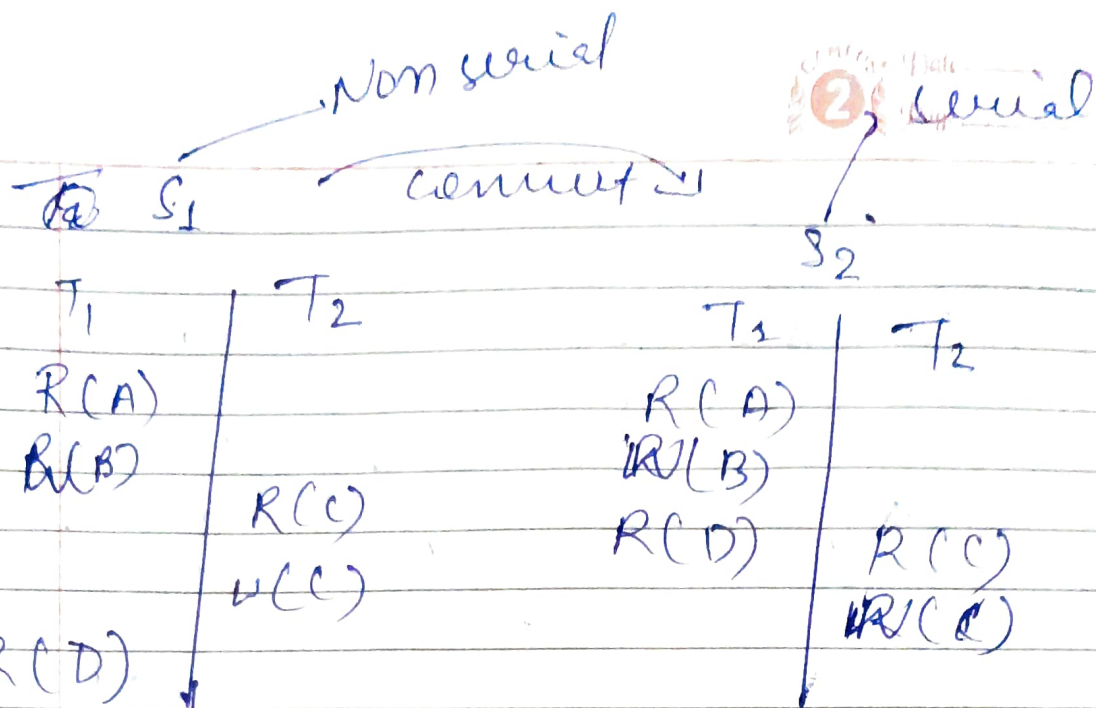
Some non serial schedules may lead to inconsistency of the database. Serializability is a concept that helps to identify non serial schedules are correct and will maintain the consistency of database.

→ What is serializable schedule?

If a given non serial schedule of n tras is equivalent to some serial schedule of n transaction then it is called as serializable schedule this is use to maintain the consistency of database. A serialization schedule helps in improving both resource utilization and CPU throughput.

Characteristics of serializable schedules.

Serializable are always
[consistent, recoverable, cascadeless, strict]

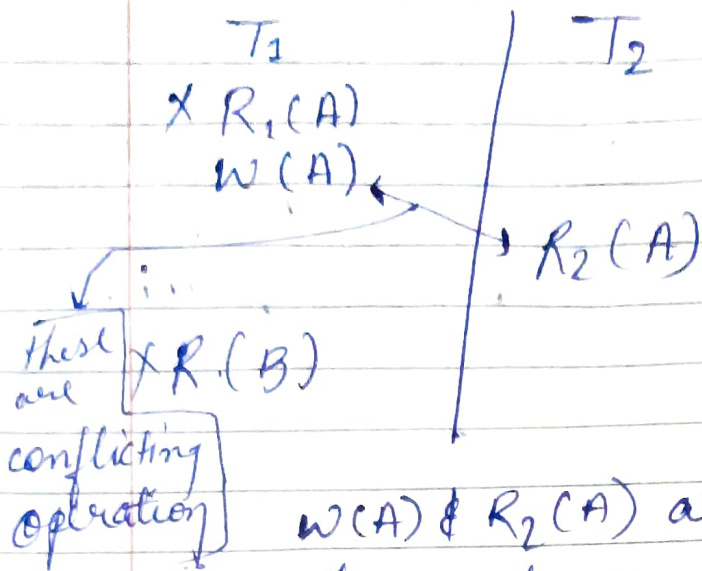


① If a given non serial sch can be converted into a serial sch by swapping its non conflicting operation then it is called as conflict serializable schedule.

What are conflicting operation?

2 operation are called as conflicting operation if the following condition holds true.

- ① Both the operation belong to different transactions.
- ② Both the operation are on same data item.
- ③ At least one of the two operation is a write operation.



$W(A)$ & $R_2(A)$ are conflicting operation the rest are not
 there can be more than 1 conflicting operation.

→ Check the given schedule is conflict serializable or not

STEP 1 -

Find & list all the ~~given~~ conflicting operation in the given schedule.

STEP 2 -

Start creating a precedence graph by drawing one node for each transaction.

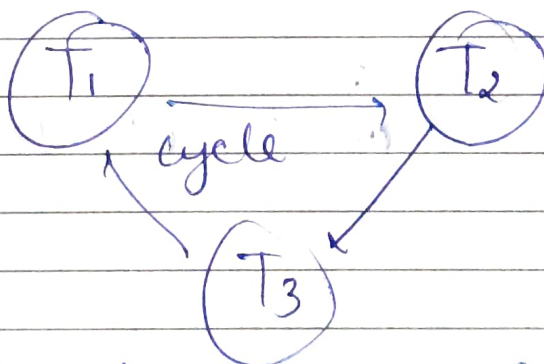
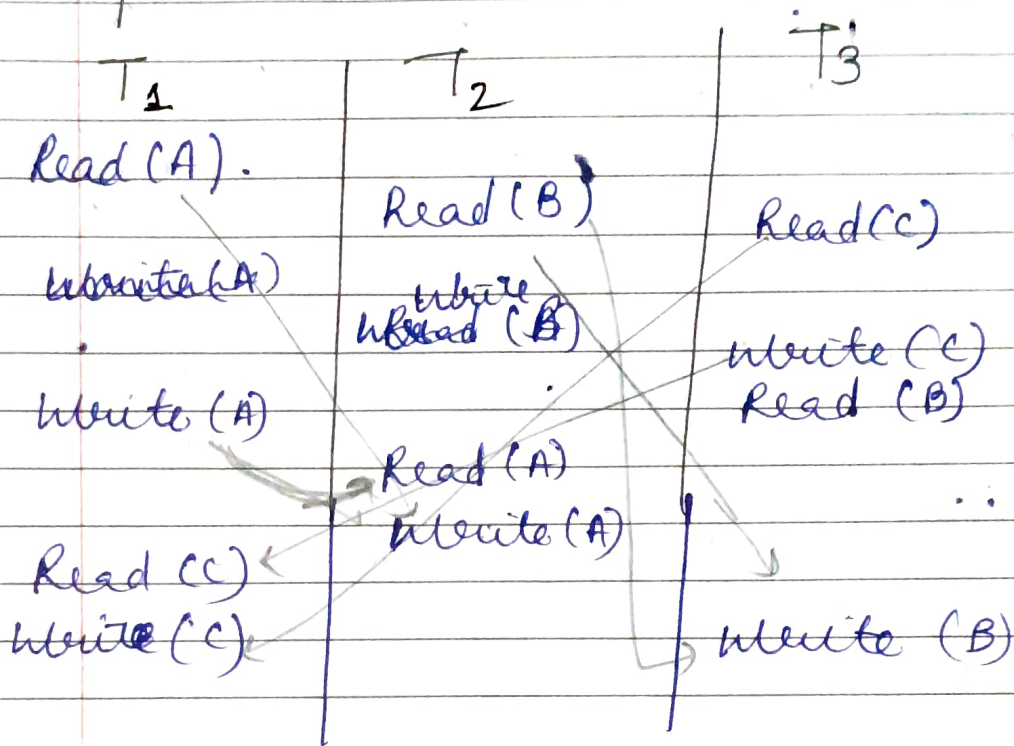
STEP 3 -

Draw an edge for each conflicting pair

STEP 4 -

Check if there is any cycle formed in the graph. If there is no cycle then, the schedule is conflict serializable.

Example -



This schedule is non conflict serializable.

Concurrency Control in DBMS

Concurrency control is a management procedure that is req for controlling.

What do you mean by concurrent execution

Multiple user system, the multiple users can access & use the same data base at one time which is known as concurrent execution of the database it means that the same database is executed on a multi user system by diff users

The problems with concurrent execution -

② In Db operation 2 main op. are read & write so there is a need to manage these op in concurrent execution of transaction

① Loss update Problem

When multiple transactions are running concurrently the lost update prob occur when the update op by one user can be over.

Loss update occurs when 2 or more op select the same row & then update the row based on the value originally selected. each transaction is unaware of the other transaction.

Transaction A	Time	Transaction B
Retrieve P	T_1	-----
-----	T_2	Retrieve B
update P.	T_3	-----
-----	T_4	update B.P.

Lost

[Both value get updated but $\rightarrow$ this updated value get lost].

① Dirty Read Problem -

DRP occurs when one transaction out of multiple transaction is allowed to see intermediate result of another transaction before it is committed.

Transaction retrieve an uncommitted update at the time T_2 . This

Update is being roll back by the transaction B at the time T_3 and thus transaction A become dependent on uncommitted change at the time T_2 .

[Solution of both the problem is commit operation.]

③ Inconsistency Analysis Prob.

This occurs when a second transaction access the same row several times and read diff data each time.

This Prob occurs when a transaction read several values from a data base while second transaction updates some of them.

* ~~Why~~ Why do we need concurrency control mechanism in DBMS, what the problem with concurrent execution

The Concurrency protocols -

- ① Atomicity
- ② Consistency
- ③ Isolation
- durability & serializability of the concurrent execution of database ~~trans~~ transactions

The 3 protocols are-

- ① ~~Same~~ Lock based CCP
- ② Time stamp concurrency control protocol
- ③ Validation based CCP